

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460705.8184 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.059
Transit depth (Rp/Rs)^2	3.92 +/- 2.32 [%]
Orbital Inclination (inc)	86.8 +/- 2.88
Airmass coefficient 1 (a1)	0.0059 +/- 0.0029
Airmass coefficient 2 (a2)	3.0 +/- 0.19
Scatter in the residuals of the lightcurve fit is	35.0 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	4.25
Transit Duration (day)	0.061 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.198 $\pm$ 0.059 vs 0.1594 (deviation 0.37 σ)
Duration fit vs published	0.061 d vs 0.0512 d (deviation 1.03 σ)
Mid-time O-C	1.1 min
Depth SNR	1.9
Residual scatter	35.0 %

Light curve

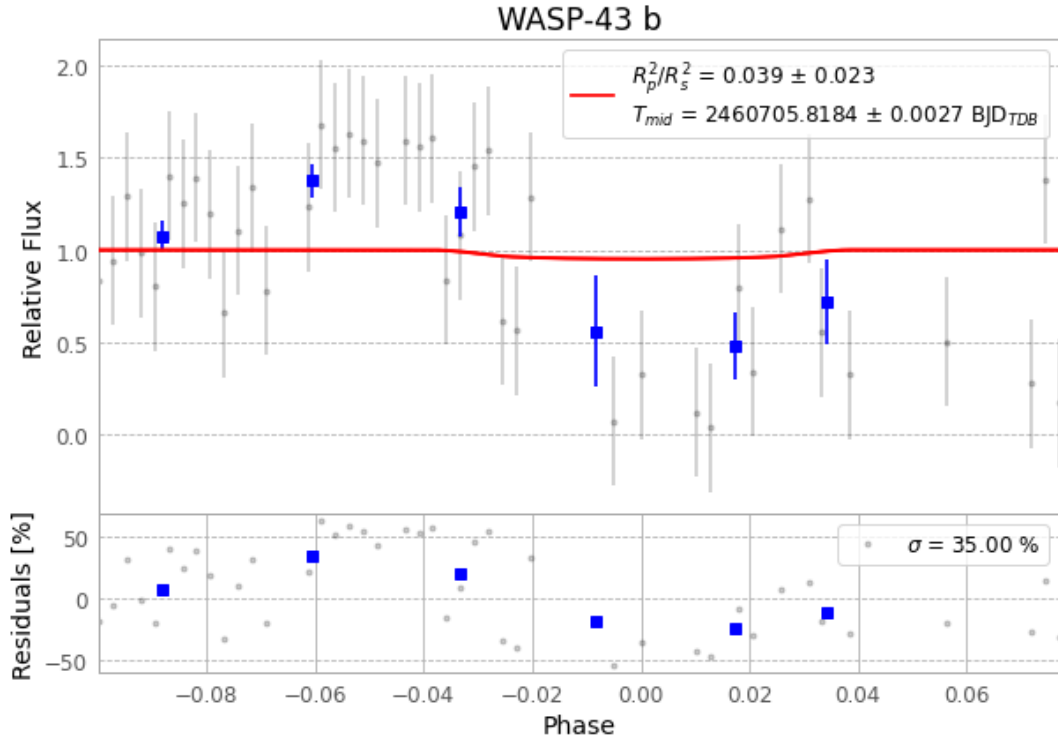


Figure 1 — FinalLightCurve_WASP-43 b_2025-01-30.png (SHA-256 6391e1d823501018...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [241, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9d17fceb12585000...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

72 FITS frames (47 MB), WASP-43250130053316.FITS → WASP-43250130090615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250130.log (2925e5210b07...), FinalLightCurve_WASP-43 b_2025-01-30.png (6391e1d82350...), FinalLightCurve_WASP-43 b_2025-01-30.csv (441f7ca241c8...)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 07:15Z.